UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

SCHOOL OF ICT

DEPARTMENT OF COMPUTER ENGINEERING

Project	Student Management System (SMS)
Module	Java in OOP Year 2 Lab 1
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Team Size	2 Students
Tech Stack	Java, Swing, SQLite, JDBC
IDE	NetBeans
Submission	GitHub Repository + Word Report
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STUDENT MANAGEMENT SYSTEM

This document presents a comprehensive report of the Student Management System. The report explains the system in a clear and practical manner, focusing on understanding, educational value, and real-world applicability rather than technical code details.

CHAPTER 1: INTRODUCTION

1.1 Project Overview

This project presents the design and implementation of a Student Management System (SMS) developed using Java. The system provides a complete solution for managing student records through a graphical interface integrated with a database. It supports secure user authentication and enables users to perform operations such as adding, updating, deleting, and searching student data efficiently.

1.2 Problem Statement

Many institutions still rely on manual or semi-digital systems for managing student records. These approaches introduce several problems:

- Data redundancy and inconsistency
- Slow search and update processes
- Lack of data validation
- Difficulty in tracking student performance
- Poor user experience

1.3 Proposed Solution

A desktop-based system was developed with a structured database, user-friendly GUI, automated validation, and efficient search and sorting features.

- Centralized database to eliminate data redundancy
- Instant search and quick updates
- Automated input validation
- Sort and filter for performance tracking
- User-friendly interface with row-click auto-fill

1.4 System Objectives

The system was designed to achieve the following objectives. These goals ensure efficient student record management and a smooth user experience.

Objective	Description
Data Management	Store and manage student records efficiently
CRUD Operations	Enable Create, Read, Update, Delete
User Authentication	Secure login and registration
Data Filtering	Filter by course and marks
Performance Tracking	Analyze student marks easily

CHAPTER 2: SYSTEM DESIGN AND ARCHITECTURE

2.1 Project Structure

The system is organized into four packages: GUI contains all user interface screens, model holds the data classes, database manages database connections and operations, and main serves as the application entry point.

Package	Description
GUI	Handles all user interfaces
Model	Business logic and data
Database	Database operations
Main	Application entry point

2.2 Object-Oriented Design

The system applies key Object-Oriented Programming principles to ensure clean, maintainable, and reusable code.

Core Components and OOP Principles:

2.3 Database Design

SQLite is used with Users and Students tables. Each student is linked to a user ensuring data isolation and security.

Core Components

Component	Type	Role
Person	Abstract Class	Base class with shared attributes
Student	Class	Handles student data and operations
User	Class	Stores logged-in user details

Database Operations	Interface	Defines CRUD methods

OOP Principles Applied

Principle	Application
Abstraction	Person class defines common structure
Inheritance	Student inherits from Person
Encapsulation	Data fields are protected/private
Polymorphism	display() behaves differently per class

CHAPTER 3: SYSTEM IMPLEMENTATION

3.1 Tools and Technologies

Tool	Purpose
Java	Core programming language for application logic
Swing	GUI framework for building user interface
JDBC	Database connectivity and query execution
SQLite	Lightweight embedded database for data storage
GitHub	Version control and team collaboration

3.2 System Modules

Includes authentication, registration, main management, search, filtering, sorting, about, and exit modules.

- **Authentication** – User login with Remember Me
- **Registration** – New account creation
- **Main Management** – Add, update, delete, view students
- **Search** – Find students by name, email, or course
- **Filter** – Filter by course or marks range
- **Sort** – Sort by name or marks
- **About** – System information display
- **Exit** – Safe application closure

3.3 Key Features

The system incorporates several features designed to enhance usability and efficiency:

Feature	Description
Row Click Auto-fill	Clicking a table row fills input fields for easy editing
Course Filter	Displays only students from selected course
Marks Filter	Slider-based filter for marks range
Sort by Name	Arranges students alphabetically
Sort by Marks	Orders students from highest to lowest marks
Search	Finds students by name, email, or course
Remember Me	Saves login for automatic next-time access

3.4 Validation Rules

Input validation ensures data integrity and prevents errors during data entry. The system checks user inputs before accepting or processing any information.

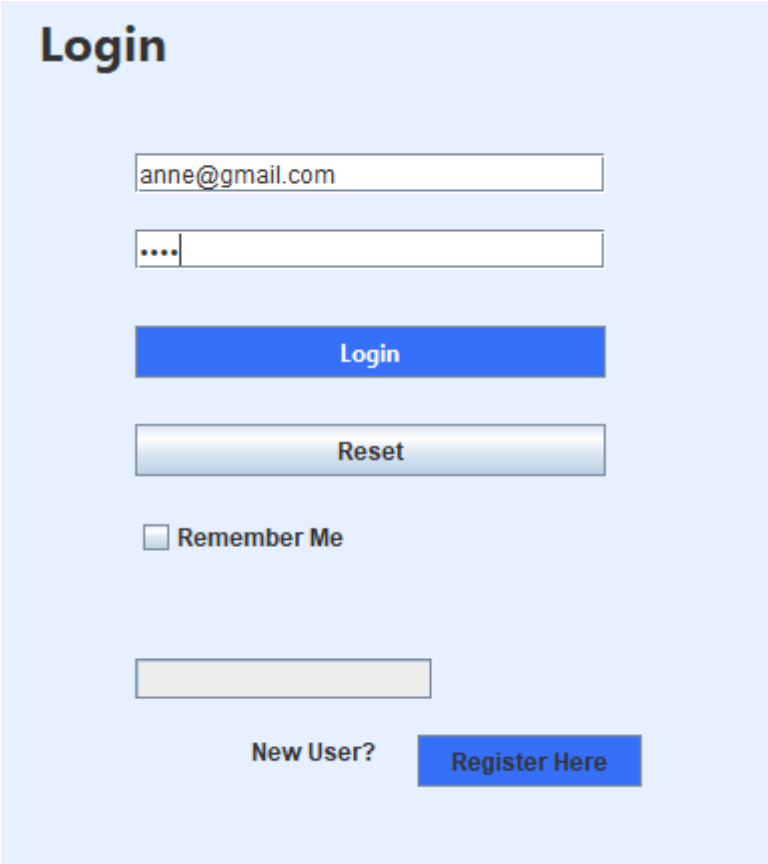
Field	Rule
Name	Must not be empty
Email	Valid format
Course	Must be selected
Marks	Must be between 0–100

CHAPTER 4: SYSTEM INTERFACE

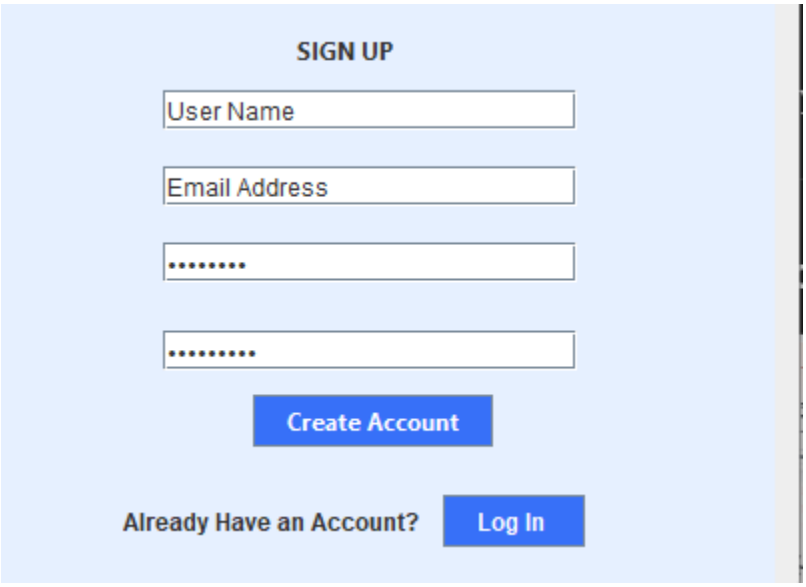
This chapter presents the graphical user interface of the Student Management System through screenshots of all major pages, with explanations of their purpose and functionality.

4.1 Login Page

The login page is the entry point to the system. It contains email and password fields for user authentication. The Login button validates credentials against the database, while the Reset button clears both fields. New users can click the Register button to create an account. The Remember Me checkbox saves login information for automatic access on future launches. A status label displays messages for successful login or errors, and a progress bar animates during the transition to the main page.

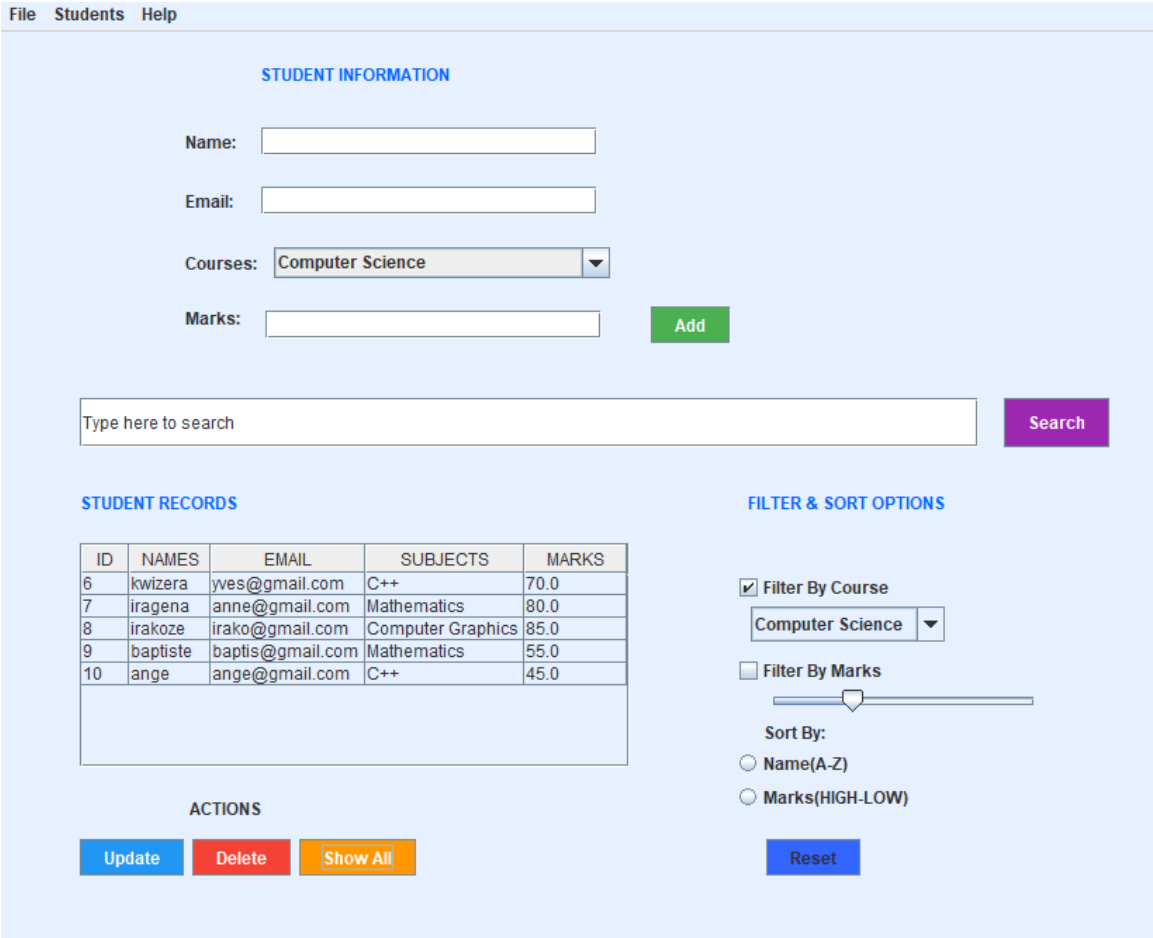


4.2 Register Page



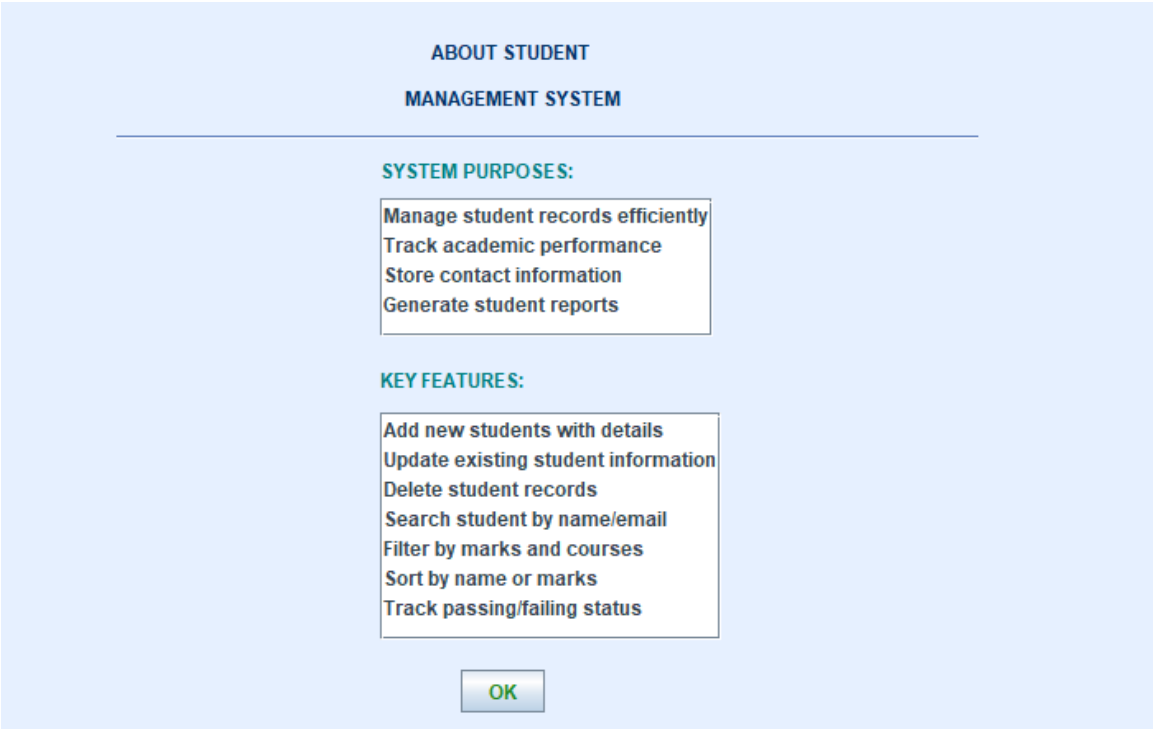
New users create accounts through this form. It includes fields for username, email, password, and password confirmation. The Create Account button validates that all fields are filled, passwords match, and email is unique before inserting the new user into the database. The Login button returns existing users to the login screen.

4.3 Main Page



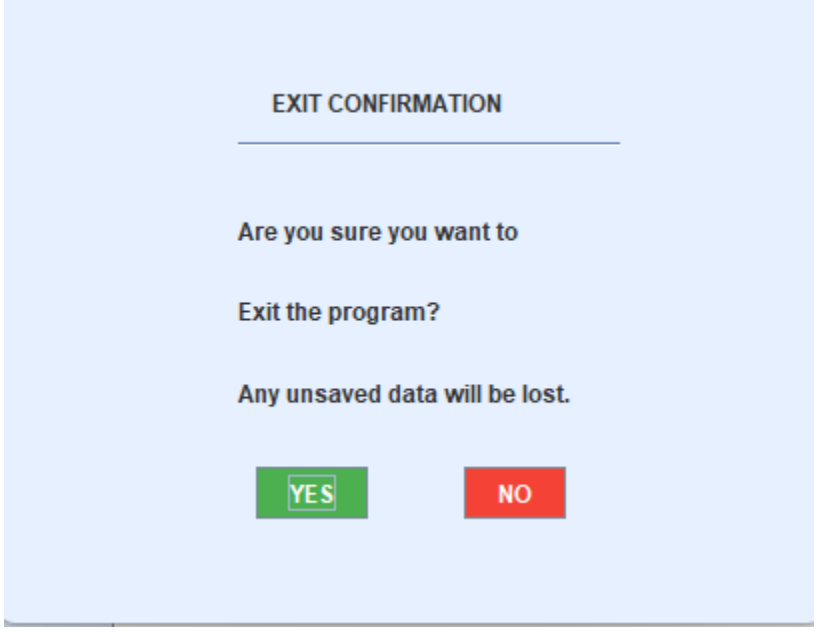
The main page is the central workspace where all student management activities occur. It is divided into several functional sections: student information input, search, records display, filters, sorting options, and action buttons. The layout is designed for efficient workflow with clear visual separation between sections.

4.4 About Page



Accessible from the Help menu, the about page displays system information including the application title, system purposes, and a complete list of key features. Users can read this information and close the dialog using the OK button.

4.5 Exit Page



When users select Exit from the File menu, this confirmation dialog appears to prevent accidental closure. It displays a warning message and provides green YES and red NO buttons. YES closes the application, while NO returns to the main page.

CHAPTER 5: CONCLUSION

The Student Management System was successfully developed using Java, providing secure authentication and complete student record management. The system features an intuitive GUI with filtering, sorting, search, and row-click auto-fill functionality. Data is securely stored in SQLite with user-specific isolation. OOP principles were effectively applied, and key challenges were resolved. The project delivers an efficient digital solution for educational institutions.

Appendix

GitHub Repository: **https://github.com/k154y/StudentManagementSystem.git**

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